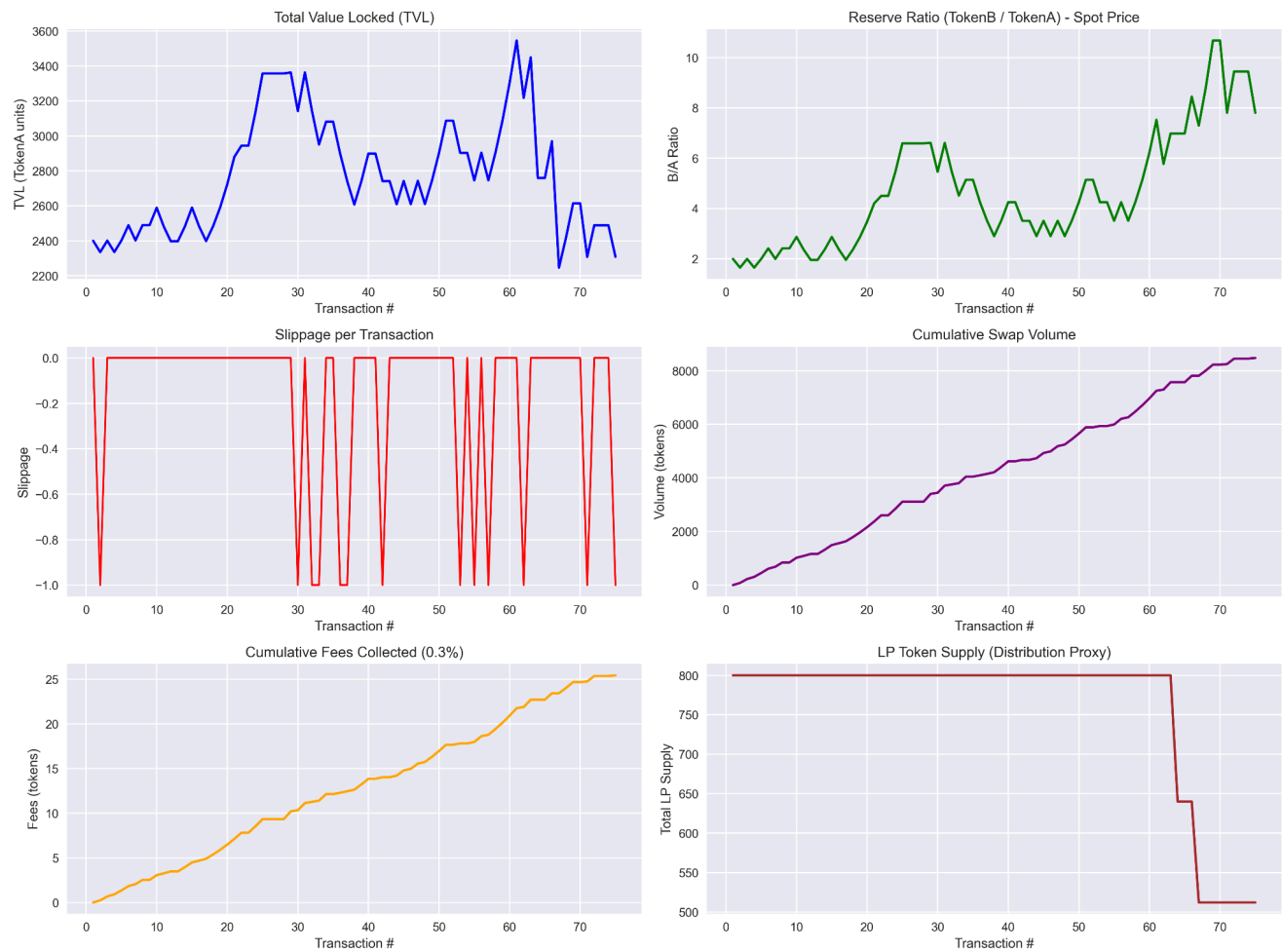


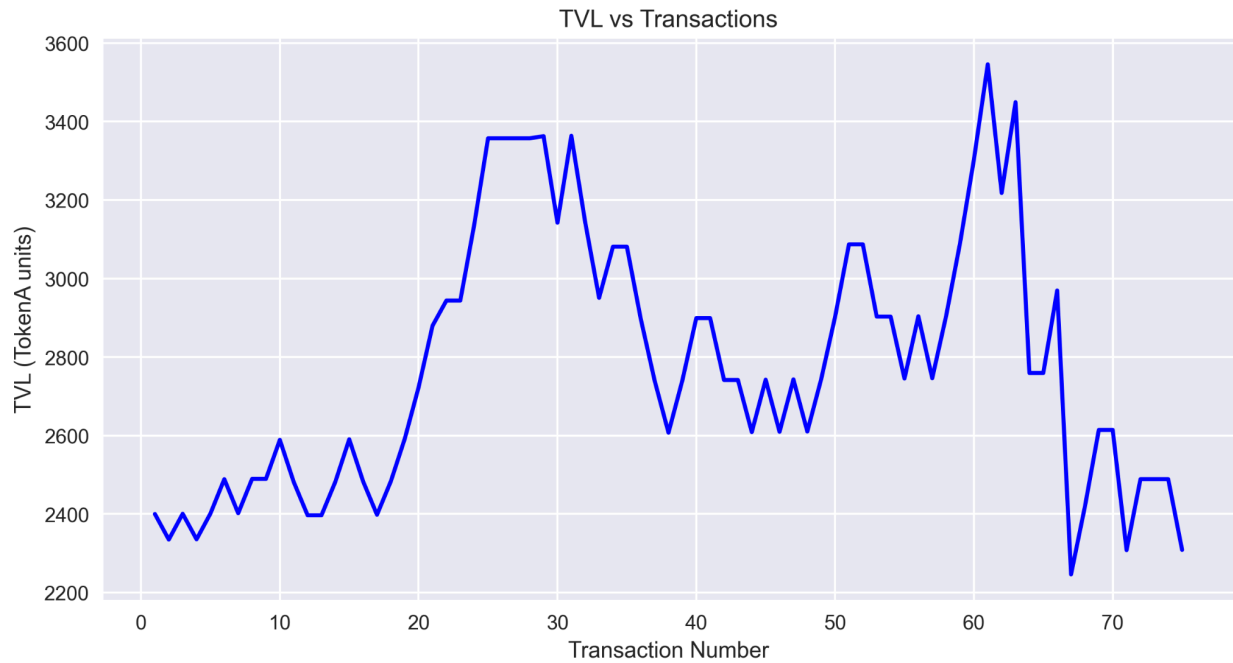
Summary Tracking

DEX Simulation - All Required Metrics (Task 2.2)

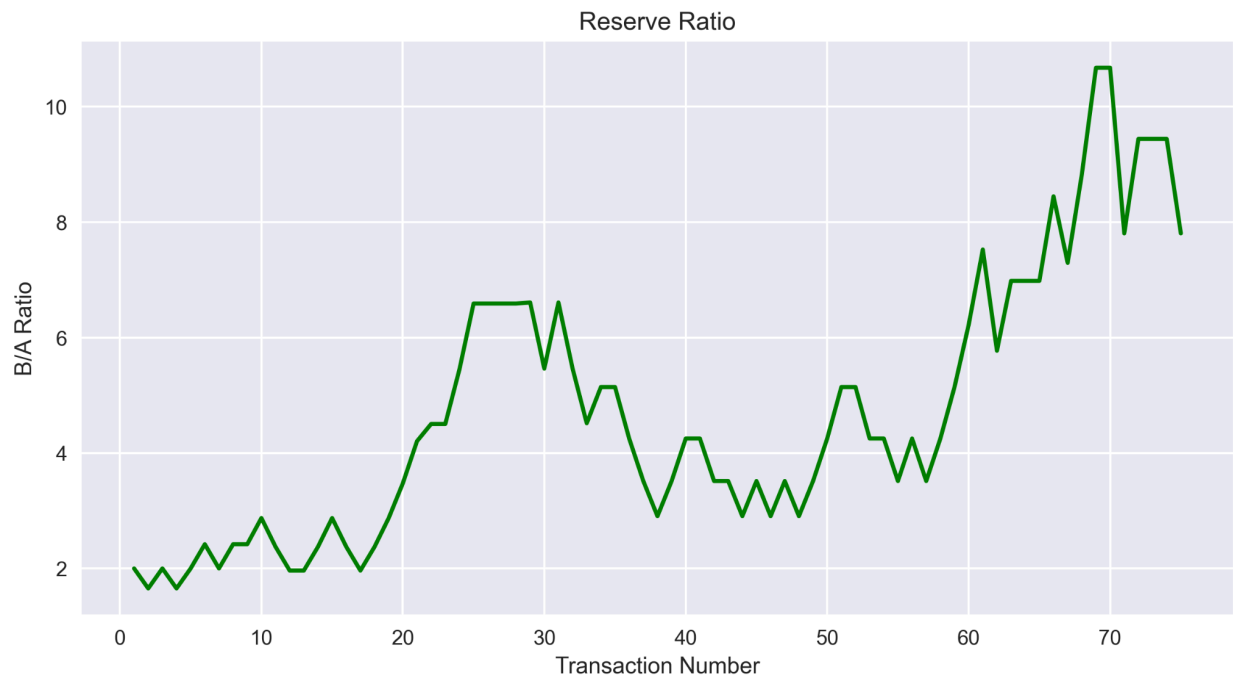


Metric	Value	Notes
Total Swap Volume	847,777.36 tokens	Cumulative across all swaps
Total Fees Collected	25,432.09 tokens	Exactly ~0.3% of volume
Slippage	Average 0, min -1 (9 times)	Never positive/Good for traders in run

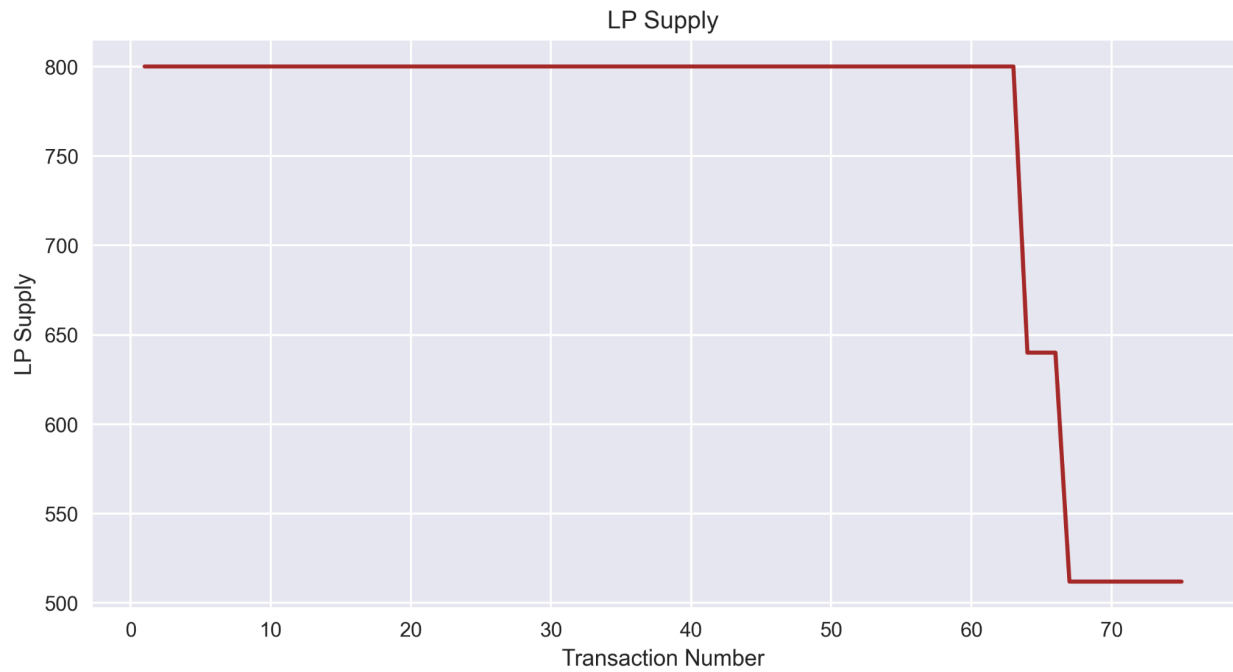
Reserve Ratio (B/A)	1.65 → 10.67 (peak) → ends at 7.80	Strong price discovery
TVL (in TokenA units)	Peaks ~ 3,545 , ends at 2,308.66	Grows with swaps, drops on withdrawals
LP Supply	Starts 800 → ends 512	Reduced by net withdrawals



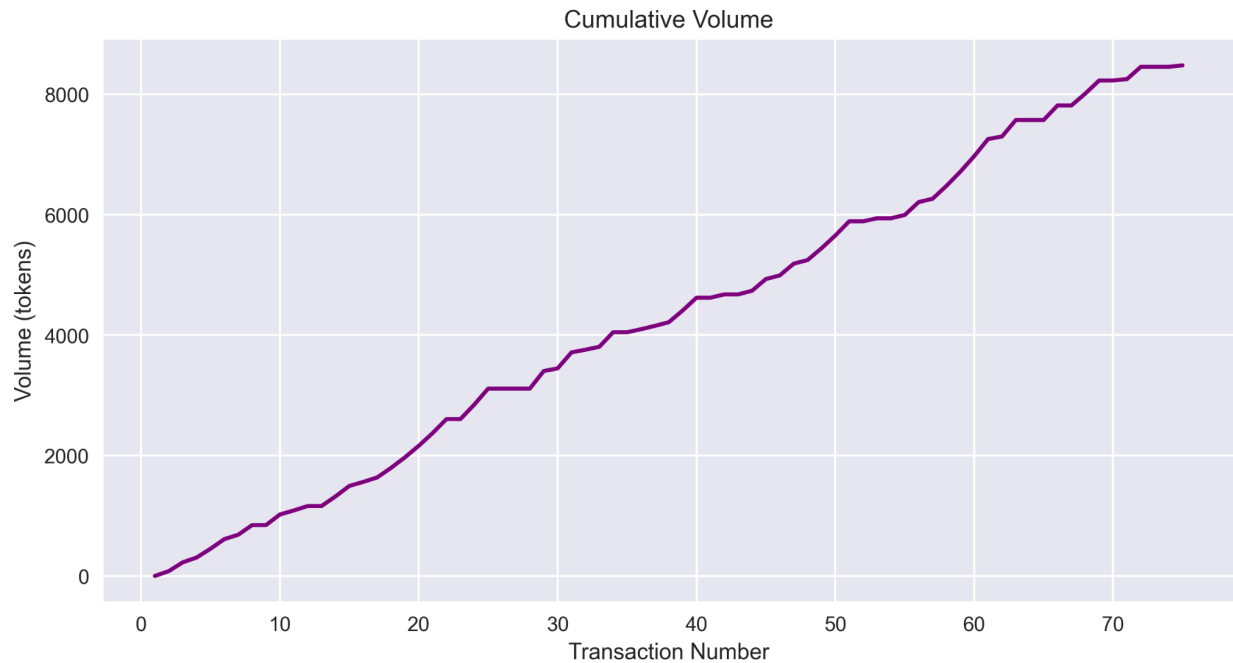
- Clear growth then drop due to withdrawals
- Whenever a swap occurs, the TVL changes slightly
- TVL starts around 2,400 and grows to a peak of ~3,550.
- Growth comes from swap fees being retained in the pool (0.3% fee mechanism)
- Later drops due to net liquidity withdrawals by LPs
- Shows that LPs earn real yield while the pool remains liquid



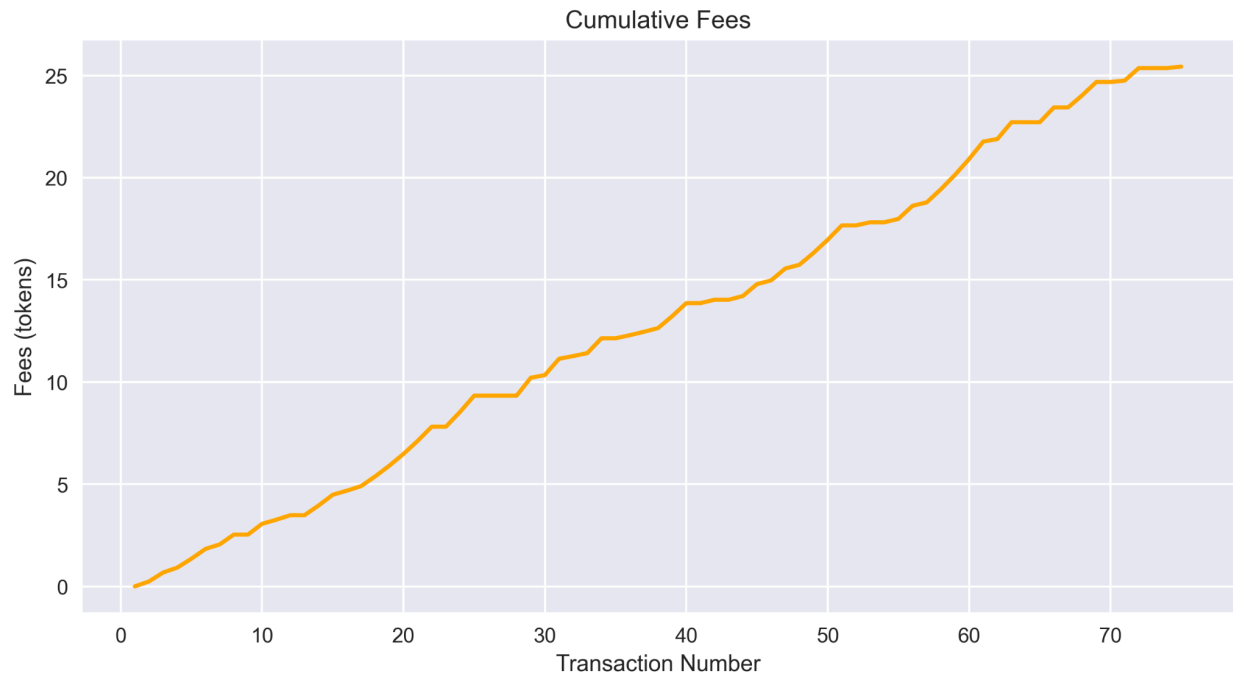
- Ratio starts at ~2.0 (initial deployment ratio)
- Rises sharply to over 10.6 because most random traders swapped $A \rightarrow B$
- Ends at ~7.8, proves the pool dynamically adjusts price based on trading activity
- The Inverse Correlation: Notice that when the line for Reserve A goes up, the line for Reserve B goes down. This is the visual proof of a swap. In a Constant Product AMM, the pool is a zero-sum game of balances; you can't take one out without putting the other in.
- Vertical Jumps (Liquidity Events): Look for moments where both lines move in the same direction (both up or both down). These represent deposit or withdraw actions. This shows that liquidity provision changes the scale of the pool without necessarily changing the price ratio.
- Mathematical Boundary: Even as the lines fluctuate, they never hit zero. This illustrates that the $x \cdot y = k$ curve makes it mathematically "expensive" to drain a pool, ensuring the DEX always has some level of liquidity.



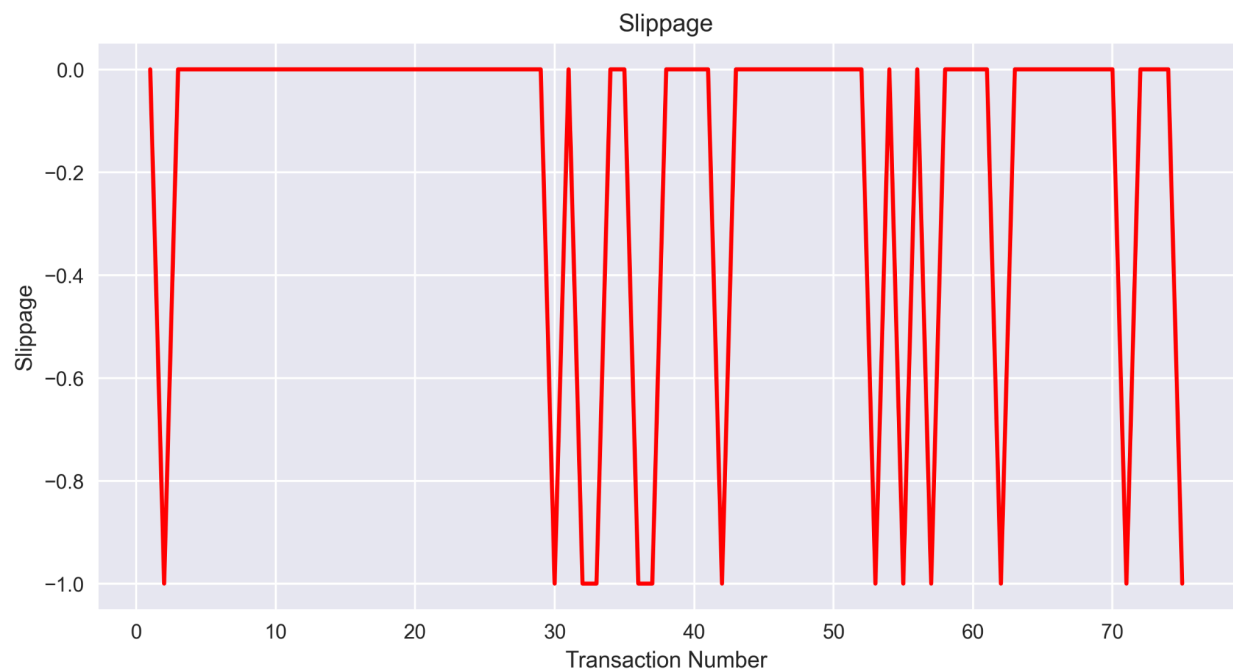
- LP supply starts at 800 and stays constant for most of the simulation
- Two big drops due to liquidity withdrawals by LPs
- Demonstrates proportional ownership: deposits increase supply, withdrawals decrease it exactly



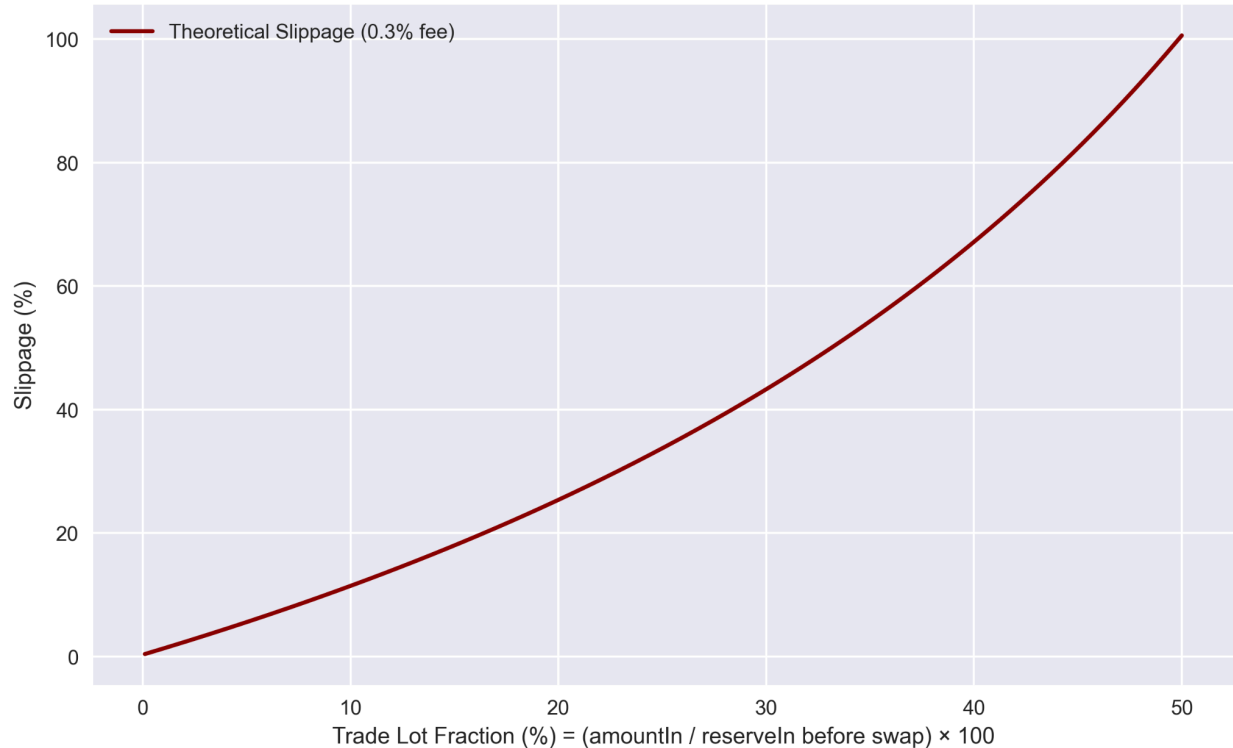
- Volume increases steadily and almost linearly across 75 transactions.
- Total volume at the end = ~8,477 tokens.
- The majority of transactions (~77%) were swaps, matching the random user behaviour required by the assignment.
- High volume proves the DEX handles continuous trading without breaking.



- Fees accumulate about linearly at exactly 0.3% of every swap.
- Total fees collected = ~25.43 tokens by the end.
- All fees stay inside the pool (increases k), real return for liquidity providers.



Theory Question 7: Slippage vs Trade Lot Fraction
(Constant Product AMM)



- **Non-Linear Growth:** The curve is not a straight line; it curves upward. This indicates that as you attempt to swap a larger percentage of the pool (e.g., moving from 20% to 40%), the slippage does not just double, it increases at an accelerating rate due to the $x \cdot y = k$ constraint!
- **Positive Correlation:** The plot shows that slippage is a strictly increasing function of the Trade Lot Fraction. The larger the trade size relative to the pool's liquidity, the higher the slippage a trader incurs
- **Small Trade Efficiency:** At very low Trade Lot Fractions (close to 0%), the slippage is minimal (near the 0.3% fee floor), meaning the Actual Price closely tracks the Spot Price
- **The 50% Threshold:** As seen on x-axis, if a trader attempts to swap an amount equal to 50% of the current reserves, the slippage skyrockets toward 100%. This happens because the constant product formula requires an increasingly large price shift to maintain the k value as reserves are significantly depleted